

SEPARATION PROCESSES II

CHE F313 · Problems and Solutions · I-Semester 2026-27

Every numerical from the uploaded slides, with its solution

Module 1 Lectures 1–6 · Module 2 Lectures 1–2

FLAG Read this first. This document mixes two kinds of statement, and they are *always* labelled:

- **SOURCE** — transcribed from the uploaded slide decks in Slides/. Questions, given data, formulas and methods. **These are the truth.**
- **DERIVED** — algebra, arithmetic and unit conversions performed here. **Generated, so verify before you rely on it.** Nothing in a **DERIVED** box overrides a **SOURCE** box.
- **VERIFY** — an independent re-computation used to check an answer.
- **TEXT** — a textbook value/formula, used *only* as secondary confirmation. Cited where used.
- **FLAG** — an ambiguity, a missing datum, or an assumption. Every place where the slides are silent, or where I had to interpret, is flagged rather than filled in silently.

If a **SOURCE** box and a **DERIVED** box ever disagree, the **SOURCE** box is wrong and the whole solution must be recomputed. Where a needed datum is absent from the slides, it is stated as absent — not invented.

Question inventory (from the uploaded decks)

ID	Source	Topic / what is asked	Solved at
Q1	Mod 1 Lec 3, slide 10	Clay catalyst: differential & cumulative analysis, A_w , $\bar{D}_s$, particle count	Q1, p. 2
Q2	Mod 1 Lec 3, slides 11–12 (table on 12; also Lec 2 slide 17)	Crushed quartz: A_w , N_w , $\bar{D}_v$, $\bar{D}_s$, $\bar{D}_w$, N_i , number fraction	Q2, p. 4
Q3	Mod 1 Lec 4, slide 19	Muller mixer: mixing index I_p and standard deviation	Q3, p. 6
Q4	Mod 1 Lec 5, slide 20	Crusher power by Rittinger's law	Q4, p. 7
Q5	Mod 2 Lec 1, slide 16 (solution graph slide 18)	Quartz screened at 10 mesh: D/F , B/F , overall effectiveness E	Q5, p. 9

Slide wording is quoted verbatim throughout, *including* the slides' own spelling and small grammar slips ("It has density of 1.2 g/cc", "analyzed calorimetrically"), so any quotation can be checked character-for-character against the deck. Two transcription notes. On Mod1-Lecture 3 slide 12 the question carries only the heading Numerical Problem, with the screen-analysis table and the shape factors inside the slide's image object; on Mod1-Lecture 5 slide 20 the powers of ten in 50×10^{-4} m and friends sit inside an image object as well. Both were read from those images and are reproduced in the **SOURCE** boxes above (the 14-row table was additionally checked row by row, see Q2). Nothing in this document was taken from a secondary summary of these slides.

REFERENCE NOTE — WHAT IS TRUSTED, AND IN WHAT ORDER

Precedence when sources disagree

1. **The uploaded decks** (Slides/, 8 files: Mod1-Lecture 1-5.pptx, Mod1-Lecture 6-compressed.pdf, Mod 2-Lecture 1-compressed.pdf, Mod 2-Lecture 2.pptx). This is the course's own material — the primary source of truth.
2. **The cited textbooks** — McCabe, Smith & Harriott, *Unit Operations of Chemical Engineering*, 7e (Ch. 28, 29), and Swain, Patra & Roy, *Mechanical Operations* (Tata McGraw Hill). Used only to confirm a standard form or a published answer.
3. **This document's **DERIVED** material** — generated arithmetic. Lowest precedence, always.

Rules applied in producing the solutions

1. The *method* used for every question is the method taught on the slides. No outside method is substituted.
2. Every number that appears in a solution came from either a **SOURCE** box or a computation in `sp2-answers-check.py` (kept alongside this file). No arithmetic was done by hand.
3. Where the slides fix a required datum (e.g. $a = 0.8$, $\Phi_s = 0.571$, ρ_p , W_i -style inputs), that value is used. Where a datum is *not* on the slides, the solution says so and either parameterises it or flags the assumption.

- SI/unit conventions follow the slides: D_p in mm with A_w in mm^2/g for the screen-analysis problems, and kW, t/h for the crushing problem.
- Answers are given with the precision the input data supports, plus an unrounded value where useful.

Q1 — SCREEN ANALYSIS OF A CLAY CATALYST

Q1

Module 1 · Lecture 3 · slide 10 · Mod1-Lecture 3.pptx

SOURCE Question as given on the slide (“Problem 1: Screen Analysis”), quoted verbatim:

“Finely divided clay is used as a catalyst in the petroleum industry. It has density of 1.2 g/cc and a sphericity of 0.5. The size analysis is as follows:”

D_{pi} / mm	0.0252	0.0178	0.0126	0.0089	0.0038
x_i (g/g)	0.088	0.178	0.293	0.194	0.247

“Show the differential & cumulative screen analysis of the mixture. Find the specific surface area and the Sauter mean diameter of the clay material. Find the number of particles in 1 g sample.”

SOURCE Method from the slides (Mod 1 Lec 3, derivations 1–13):

$$A_w = \frac{6}{\rho_p \Phi_s} \sum_{i=1}^n \frac{x_i}{D_{pi}}, \quad \bar{D}_s = \frac{1}{\sum_i x_i / D_{pi}} = \frac{6}{\Phi_s \rho_p A_w}, \quad N_w = \frac{1}{a \rho_p} \sum_{i=1}^n \frac{x_i}{D_{pi}^3}$$

Cumulative analysis: add the increments consecutively starting from the smallest particles, and plot the sums against the maximum particle diameter of the increment (Lec 2 slide 16).

Step 1 — differential and cumulative analysis

DERIVED The differential analysis is the tabulated x_i versus D_{pi} already given. The cumulative “fraction smaller than D_{pi} ” is the running sum from the smallest size; “fraction larger” is its complement. Check of the given data: $\sum x_i = 0.088 + 0.178 + 0.293 + 0.194 + 0.247 = 1.000$.

D_{pi} / mm	x_i (differential)	cumulative smaller	cumulative larger
0.0038	0.247	0.2470	0.7530
0.0089	0.194	0.4410	0.5590
0.0126	0.293	0.7340	0.2660
0.0178	0.178	0.9120	0.0880
0.0252	0.088	1.0000	0.0000

DERIVED Rows are listed from the smallest increment upward, which is the order used when building a cumulative distribution.

Step 2 — specific surface area and Sauter mean diameter

DERIVED Unit note: D_{pi} is in mm, so ρ_p must be in g/mm^3 :

$$\begin{aligned} \rho_p &= 1.2 \text{ g}/\text{cm}^3 = 1.2 \times 10^{-3} \text{ g}/\text{mm}^3 \\ \sum_i \frac{x_i}{D_{pi}} &= 123.544 \text{ mm}^{-1} \quad (\text{e.g. } 0.088/0.0252 = 3.4921, \quad 0.247/0.0038 = 65.000) \\ A_w &= \frac{6}{\rho_p \Phi_s} \sum_i \frac{x_i}{D_{pi}} = \frac{6}{(1.2 \times 10^{-3})(0.5)} (123.544) = 1.2354 \times 10^6 \text{ mm}^2/\text{g} \\ \bar{D}_s &= \frac{1}{\sum_i x_i / D_{pi}} = \frac{1}{123.544} = 0.008094 \text{ mm} = 8.09 \text{ }\mu\text{m} \end{aligned}$$

VERIFY Consistency of the two slide formulas: $\frac{6}{\Phi_s \rho_p A_w} = \frac{6}{(0.5)(1.2 \times 10^{-3})(1.2354 \times 10^6)} = 0.008094 \text{ mm} = \bar{D}_s$
 ✓
 Sanity: $A_w = 1.2354 \times 10^6 \text{ mm}^2/\text{g} = 1.24 \text{ m}^2/\text{g}$, which is the right order for a powder of $\sim 8 \text{ }\mu\text{m}$ particles.

Step 3 — number of particles in a 1 g sample

DERIVED For the same data,

$$\sum_i \frac{x_i}{D_{pi}^3} = 4.9601 \times 10^6 \text{ mm}^{-3}, \quad N_w = \frac{1}{a\rho_p} \sum_i \frac{x_i}{D_{pi}^3} = \frac{4.9601 \times 10^6}{a(1.2 \times 10^{-3})} = \frac{4.1334 \times 10^9}{a} \text{ g}^{-1}$$

The slide does not state the volume shape factor a , so the result is left in terms of it. For the sphere value used elsewhere in the course ($a = \pi/6 = 0.5236$, Lec 3 slide 6 and the textbook),

$$N_w = \frac{4.1334 \times 10^9}{0.5236} = 7.89 \times 10^9 \text{ particles/g}$$

FLAG a is *not* given on this slide (unlike Q2, where $a = 0.8$ is stated). The value $7.9 \times 10^9/\text{g}$ assumes $a = \pi/6$ (spheres). Since $N_w \propto 1/a$, quoting a from the question sheet changes the answer: $a = 0.5 \rightarrow 8.3 \times 10^9/\text{g}$, $a = 0.8 \rightarrow 5.2 \times 10^9/\text{g}$. Confirm the intended a before submitting.

Answers (Q1).

(a) Differential and cumulative tables as above ($\sum x_i = 1.000$).

(b) $A_w = 1.2354 \times 10^6 \text{ mm}^2/\text{g}$ ($\approx 1.24 \text{ m}^2/\text{g}$); $\bar{D}_s = 0.00809 \text{ mm} = 8.09 \text{ }\mu\text{m}$.

(c) $N_w = (4.133 \times 10^9)/a$ particles per gram; with $a = \pi/6$: $\approx 7.9 \times 10^9$ particles/g. (*a not given on the slide — see the flag.*)

Q2 — SCREEN ANALYSIS OF CRUSHED QUARTZ

Q2

Module 1 · Lecture 3 · slides 11–12 (table on slide 12) · also Lec 2 slide 17

SOURCE Question as given on the slide (“Problem 2: Screen Analysis”), quoted verbatim:

“The screen analysis shown in Table applies to a sample of crushed quartz. The density of the particles is 2650 kg/m³ (0.00265 g/mm³), and the shape factors are $a = 0.8$ and $\Phi_s = 0.571$. For the material between 4-mesh and 200-mesh in particle size, calculate (a) A_w in square millimeters per gram and N_w in particles per gram, (b) D_v (c) D_s (d) D_w and (e) N_i for the 150/200-mesh increment. (f) What fraction of the total number of particles is in the 150/200-mesh increment?”

SOURCE Data table as printed on slide 12 (the slide’s only text is the heading **Numerical Problem**; the same table also appears on Lecture 2 slide 17). Columns as on the slide: mesh, screen opening, mass fraction retained x_i , average particle diameter in increment, cumulative fraction smaller than D_{pi} .

Mesh	Screen opening D_{pi} /mm	retained x_i	D_{pi} in increment/mm	cumulative smaller
4	4.699	0.0000	—	1.0000
6	3.327	0.0251	4.013	0.9749
8	2.362	0.1250	2.845	0.8499
10	1.651	0.3207	2.007	0.5292
14	1.168	0.2570	1.409	0.2722
20	0.833	0.1590	1.001	0.1132
28	0.589	0.0538	0.711	0.0594
35	0.417	0.0210	0.503	0.0384
48	0.295	0.0102	0.356	0.0282
65	0.208	0.0077	0.252	0.0205
100	0.147	0.0058	0.178	0.0147
150	0.104	0.0041	0.126	0.0106
200	0.074	0.0031	0.089	0.0075
Pan	—	0.0075	0.037	0.0000

DERIVED Internal check of the table: each increment diameter is the arithmetic mean of the two defining screen openings, e.g. $(4.699 + 3.327)/2 = 4.013$, $(3.327 + 2.362)/2 = 2.845$, ..., $(0.074 + 0)/2 = 0.037$ for the pan. All 13 values reproduce the table to within 0.0005 mm (the table prints three decimals, so e.g. the 14-mesh mean 1.4095 is printed as 1.409); the data is therefore self-consistent and no reading error is involved. The 12 increments from 4-mesh to 200-mesh carry $\sum x_i = 0.9925$; the remaining 0.0075 is the pan. Because the question says “between 4-mesh and 200-mesh”, the pan is *excluded* from the answer (the textbook solution does the same) — see the flag below for the numbers if it is included.

SOURCE Method from the slides:

$$A_w = \frac{6}{\rho_p \Phi_s} \sum \frac{x_i}{D_{pi}}, \quad N_w = \frac{1}{a \rho_p} \sum \frac{x_i}{D_{pi}^3}, \quad \bar{D}_v = \left[\frac{1}{\sum x_i / D_{pi}^3} \right]^{1/3},$$

$$\bar{D}_s = \frac{1}{\sum x_i / D_{pi}}, \quad \bar{D}_w = \sum x_i D_{pi}, \quad N_i = \frac{m_i}{a \rho_p D_{pi}^3} = \frac{x_i}{a \rho_p \bar{D}_{pi}^3} \text{ (per unit mass)}$$

Step 1 — the two sums (12 increments, 4-mesh to 200-mesh)

DERIVED

$$\sum_{i=1}^{12} \frac{x_i}{D_{pi}} = 0.82780 \text{ mm}^{-1} \quad \sum_{i=1}^{12} \frac{x_i}{D_{pi}^3} = 8.7932 \text{ mm}^{-3}$$

work for the first increment, $D_{pi} = 4.013$ and $x_i = 0.0251$: $0.0251/4.013 = 0.00626$ and $0.0251/4.013^3 = 0.000389$.

Step 2 — (a) specific surface area and number of particles

DERIVED

$$A_w = \frac{6}{\rho_p \Phi_s} \sum \frac{x_i}{D_{pi}} = \frac{6}{(2.65 \times 10^{-3})(0.571)} (0.82780) = 3282 \text{ mm}^2/\text{g} = 32.8 \text{ cm}^2/\text{g}$$

$$N_w = \frac{1}{a \rho_p} \sum \frac{x_i}{D_{pi}^3} = \frac{8.7932}{(0.8)(2.65 \times 10^{-3})} = 4148 \text{ particles/g}$$

Step 3 — (b) volume mean, (c) Sauter, (d) mass mean diameters

DERIVED

$$\begin{aligned}\bar{D}_v &= \left[\frac{1}{\sum x_i/D_{pi}^3} \right]^{1/3} = \left[\frac{1}{8.7932} \right]^{1/3} = 0.4845 \text{ mm} \\ \bar{D}_s &= \frac{1}{\sum x_i/D_{pi}} = \frac{1}{0.82780} = 1.2080 \text{ mm} \\ \bar{D}_w &= \sum x_i D_{pi} = 0.0251(4.013) + 0.1250(2.845) + \cdots + 0.0031(0.089) = 1.6775 \text{ mm}\end{aligned}$$

Step 4 — (e) particles in the 150/200-mesh increment

DERIVED For that increment $x_i = 0.0031$ and $D_{pi} = 0.089 \text{ mm}$ (per unit mass of the whole sample):

$$N_i = \frac{x_i}{a\rho_p D_{pi}^3} = \frac{0.0031}{(0.8)(2.65 \times 10^{-3})(0.089)^3} = 2074 \text{ particles/g}$$

Step 5 — (f) fraction of the total number of particles

DERIVED The fraction of all particles that lie in that increment is its share of $\sum x_i/D_{pi}^3$:

$$\frac{N_i}{N_T} = \frac{x_i/D_{pi}^3}{\sum_j x_j/D_{pj}^3} = \frac{0.0031/0.089^3}{8.7932} = 0.5001 \approx 50\%$$

Note the contrast with its *mass* fraction, $x_i = 0.0031 = 0.31\%$: half of all the particles sit in this one fine increment even though it is a negligible part of the mass.

VERIFY Independent checks: (i) the algebra route $N_i/N_T = [(1/(a\rho_p))(x_i/D_{pi}^3)]/N_w = 2074/4148 = 0.500$ agrees; (ii) the published worked solution of this textbook problem, as reproduced in course material, gives $\sum x_i/D_{pi} = 0.8278$, $\sum x_i/D_{pi}^3 = 8.7932$, $A_w = 3282 \text{ mm}^2/\text{g}$, $N_w = 4148 \text{ particles/g}$, $N_i = 2074$ and a fraction of 50% **TEXT** — i.e. this solution reproduces the published answer to the digits quoted, which is strong evidence that the table was transcribed correctly.

FLAG Two points to confirm with the instructor:

- Scope.** “Between 4-mesh and 200-mesh” excludes the pan, so $\sum x_i = 0.9925$ and the denominators are not normalised to 1. Including the pan gives different numbers ($A_w = 4086 \text{ mm}^2/\text{g}$, $N_w = 73\,990 \text{ particles/g}$, $\bar{D}_v = 0.185 \text{ mm}$, $\bar{D}_s = 0.970 \text{ mm}$, $\bar{D}_w = 1.678 \text{ mm}$, $N_i/N_T = 2.8\%$). The exclusion is what the textbook solution does, so it is used above.
- Which diameter to use.** The solution uses the increment-mean column D_{pi} from the table. The 150/200 value there is 0.089 mm (not 0.104 or 0.074, the screen openings).

Answers (Q2).

- $A_w = 3282 \text{ mm}^2/\text{g}$, $N_w = 4148 \text{ particles/g}$.
- $\bar{D}_v = 0.484 \text{ mm}$.
- $\bar{D}_s = 1.208 \text{ mm}$.
- $\bar{D}_w = 1.677 \text{ mm}$.
- $N_i = 2074 \text{ particles/g}$ (150/200-mesh increment).
- $N_i/N_T = 0.500$ — about **50%** of all the particles, against only 0.31% of the mass.

Q3 — MIXING INDEX IN A MULLER MIXER

Q3

Module 1 · Lecture 4 · slide 19 · Mod1-Lecture 4.pptx

SOURCE Question as given on the slide (“Problem 1”), quoted verbatim:

“A silty soil containing 14 percent moisture was mixed in a large muller mixer with 10 wt% of a tracer consisting of dextrose and picric acid. After 3 min of mixing, 12 random samples were taken from the mix and analyzed calorimetrically for tracer material. The measured concentrations in the sample were in weight percent tracer, 10.24, 9.30, 7.94, 10.24, 11.08, 10.03, 11.91, 9.72, 9.20, 10.76, 10.97, 10.55. Calculate the mixing index IP and the standard deviation.”

SOURCE Method from the slides (Mod 1 Lec 4, cohesive solids / pastes — and Lec 4 slide 16 for s):

$$s = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}} = \sqrt{\frac{\sum_{i=1}^N x_i^2 - \bar{x} \sum_{i=1}^N x_i}{N - 1}}, \quad \sigma_0 = \sqrt{\mu(1 - \mu)}, \quad I_s = \frac{\sigma_0}{s}$$

“For cohesive solids, instead of number, mass fraction is used”; x_i = mass fraction of A in each sample, $\bar{x}$ = average of the mass fractions, N = number of spot samples; σ_0 is the theoretical standard deviation at zero mixing. The soil is wet and cohesive, and the samples were analysed for tracer *mass* percent, so the paste equations apply.

Step 1 — standard deviation of the samples

DERIVED

$$\bar{x} = \frac{1}{12} \sum_{i=1}^{12} x_i = \frac{121.94}{12} = 10.1617 \text{ wt\%} \quad \sum_{i=1}^{12} (x_i - \bar{x})^2 = 11.890 \text{ (wt\%)}^2$$

$$s = \sqrt{\frac{11.890}{12 - 1}} = 1.0397 \text{ wt\%} = 0.010397 \text{ (mass fraction)}$$

(The N instead of $N - 1$ denominator would give $s = 0.9954 \text{ wt\%} = 0.009954$ as a fraction.)

Step 2 — theoretical standard deviation at zero mixing

DERIVED The mixture was charged with 10 wt% tracer, so $\mu = 0.10$:

$$\sigma_0 = \sqrt{\mu(1 - \mu)} = \sqrt{0.10 \times 0.90} = \sqrt{0.09} = 0.30$$

Step 3 — mixing index

DERIVED

$$I_p = \frac{\sigma_0}{s} = \frac{0.30}{0.010397} = 28.9$$

Since s has fallen well below σ_0 , the value of I_p strongly above 1 indicates that the mixer has mixed the tracer far more uniformly than a completely unmixed, segregated charge would be (compare $I_s = \sigma_e/s$ for granular solids, which rises from 0 towards 1 as mixing proceeds and is capped near 1 for a random mixture).

FLAG Four judgement calls, all of which change the last digit or more:

1. **μ : charged or measured?** $\mu = 0.10$ is used, because that is the tracer actually charged (“mixed with 10 wt% of a tracer”). Using the measured mean instead, $\mu = \bar{x} = 0.1016$, gives $\sigma_0 = 0.3022$ and $I_p = 29.1$.
2. **$N - 1$ or N ?** The slide’s formula is $s = \sqrt{\sum (x_i - \bar{x})^2 / (N - 1)}$, used above. Some textbooks use N : that gives $s = 0.995\%$ and $I_p = 30.1$.
3. **Number vs mass fraction.** The question says “silty soil” (cohesive, 14% moisture) and reports wt% tracer, so the paste form $\sigma_0 = \sqrt{\mu(1 - \mu)}$ with mass fractions is used. The granular form $\sigma_e = \sqrt{\mu_p(1 - \mu_p)/n}$ needs n , the number of particles per sample, which the slide does not give.
4. **Symbol.** The slide writes the paste mixing index as I_s ; the question calls it I_p . They are the same quantity.

Answers (Q3).

$\bar{x} = 10.16 \text{ wt\%}$, $s = 1.04 \text{ wt\%}$ (i.e. 0.0104 as a mass fraction; $s = 0.995 \text{ wt\%}$ if the N denominator is used), $\sigma_0 = 0.30$, $I_p = 0.30/0.0104 = 28.9$ (≈ 29).

With the measured mean as μ and the N -denominator, the same working gives $I_p \approx 30$.

Q4 — CRUSHER POWER BY RITTINGER'S LAW

Q4

Module 1 · Lecture 5 · slide 20 · Mod1-Lecture 5.pptx

SOURCE Question as given on the slide (“Problem 1: Size reduction”), quoted verbatim:

“Particles of the average feed size of 50×10^{-4} m are crushed to an average product size of 10×10^{-4} m at the rate of 20 tonnes per hour. At this rate, the crusher consumes 40 kW of power which 5 kW are required for running the mill empty. Calculate the power consumption if 12 tonnes/h of this product is crushed to 5×10^{-4} m size in the same mill? Assume Rittinger's Law is applicable.”

SOURCE Method fixed by the slide (Mod 1 Lec 5, Rittinger's law):

$$\frac{P}{\dot{m}} = K_R \left[\frac{1}{D_{sb}} - \frac{1}{D_{sa}} \right]$$

with D_{sa} = feed (a) size and D_{sb} = product (b) size. The question directs Rittinger's law, so it is used — not Kick's or Bond's.

Step 1 — useful (net) power of the first duty

DERIVED The 5 kW is spent running the mill empty, so the power actually doing crushing is

$$P_{\text{net},1} = 40 - 5 = 35 \text{ kW at } \dot{m}_1 = 20 \text{ t/h}$$

$$\frac{P_{\text{net},1}}{\dot{m}_1} = \frac{35}{20} = 1.75 \text{ kW h/t}$$

Step 2 — the Rittinger constant of this mill and material

DERIVED With $D_{sa} = 50 \times 10^{-4}$ m and $D_{sb} = 10 \times 10^{-4}$ m,

$$\frac{1}{D_{sb}} - \frac{1}{D_{sa}} = \frac{1}{10 \times 10^{-4}} - \frac{1}{50 \times 10^{-4}} = 1000 - 200 = 800/\text{m}$$

$$K_R = \frac{P/\dot{m}}{[1/D_{sb} - 1/D_{sa}]} = \frac{1.75}{800} = 2.1875 \times 10^{-3} \text{ kW h m/t}$$

Step 3 — the second duty

DERIVED Now the same mill crushes *this product* ($D_{sa} = 10 \times 10^{-4}$ m) down to 5×10^{-4} m at $\dot{m}_2 = 12$ t/h:

$$\frac{1}{D_{sb}} - \frac{1}{D_{sa}} = \frac{1}{5 \times 10^{-4}} - \frac{1}{10 \times 10^{-4}} = 2000 - 1000 = 1000/\text{m}$$

$$P_{\text{net},2} = K_R \dot{m}_2 \left[\frac{1}{D_{sb}} - \frac{1}{D_{sa}} \right] = (2.1875 \times 10^{-3})(12)(1000) = 26.25 \text{ kW}$$

Restoring the fixed empty-mill load:

$$P_2 = P_{\text{net},2} + 5 = 31.25 \text{ kW}$$

Step 4 — the same calculation done entirely with power ratios

DERIVED A second route, keeping the mill load in the ratio:

$$\frac{P_2 - 5}{P_1 - 5} = \frac{\dot{m}_2}{\dot{m}_1} \cdot \frac{[1/D_{sb,2} - 1/D_{sa,2}]}{[1/D_{sb,1} - 1/D_{sa,1}]} = \frac{12}{20} \cdot \frac{1000}{800} = 0.75$$

$$P_2 = 5 + 0.75(40 - 5) = 5 + 26.25 = 31.25 \text{ kW}$$

VERIFY The two routes agree exactly ($26.25 + 5 = 31.25$), and both are dimensionally consistent: kW h/t $\times$ t/h $\times$ 1/m must be checked with D in metres, giving kW. Note also that the second duty crushes a *finer* feed ($10 \rightarrow 5$ instead of $50 \rightarrow 10$), and the size-reduction step $(1/5 - 1/10) = 1000$ exceeds the first duty's $(1/10 - 1/50) = 800$ per metre, so the smaller throughput does not make the power small.

FLAG Two readings of “power consumption”, both shown:

- **Total power drawn by the crusher: 31.25 kW** — the answer above, because the question asks for the *power consumption* of the machine and the empty-mill load is still drawn.
- **Net crushing power: 26.25 kW** — if only the work of size reduction is wanted. Quoting 26.25 kW as the machine’s consumption would double-count the treatment of the 5 kW already handled in the first duty.

Answers (Q4). $K_R = 2.1875 \times 10^{-3} \text{ kW h m/t}$; $P_{\text{net},2} = 26.25 \text{ kW}$; $P_2 = 31.25 \text{ kW}$ including the empty-mill power.

Q5 — SCREEN EFFECTIVENESS OF A QUARTZ MIXTURE

Q5

Module 2 · Lecture 1 · slide 16, solution graph slide 18 · Mod 2-Lecture 1-compressed.pdf

SOURCE Question as given on the slide (“Problem on Screen Effectiveness”), quoted verbatim:

“A quartz mixture having the screen analysis shown in Table is screened through a standard 10-mesh screen. The cumulative screen analysis of overflow and underflow are given in Table. Calculate — the mass ratios of the overflow and underflow to feed and — the overall effectiveness of the screen.”

SOURCE Data table as printed on slide 16 — the slide’s caption is “Screen analyses — Cumulative fraction LARGER than D_p ”, with columns Feed, Overflow, Underflow:

Mesh	D_p/mm	Feed x_F	Overflow x_D	Underflow x_B
4	4.699	0	0	
6	3.327	0.025	0.071	
8	2.362	0.15	0.43	0
10	1.651	0.47	0.85	0.195
14	1.168	0.73	0.97	0.58
20	0.833	0.885	0.99	0.83
28	0.589	0.94	1.00	0.91
35	0.417	0.96		0.94
65	0.208	0.98		0.975
Pan		1.00		1.00

DERIVED Reading of the data: the cut diameter is the 10-mesh opening, $D_{pc} = 1.651 \text{ mm}$ (the slide says the test screen is 10-mesh), so the table row for 10 mesh supplies the three values needed directly — $x_F = 0.47$, $x_D = 0.85$, $x_B = 0.195$. No interpolation is needed, because the slide’s table is already tabulated at that size. Blank cells in the overflow column (35 and 65 mesh) mean that no overflow stream was analysed above 28 mesh; they are not needed for the calculation, which uses only the 10-mesh row.

SOURCE Method from the slides (Mod 2 Lec 1, hand-worked slides 12–15):

Overall and oversize balances give the split,

$$\frac{D}{F} = \frac{x_F - x_B}{x_D - x_B}, \quad \frac{B}{F} = \frac{x_F - x_D}{x_B - x_D}$$

and the effectiveness of the screen in each direction,

$$E_A = \frac{D x_D}{F x_F}, \quad E_B = \frac{B(1 - x_B)}{F(1 - x_F)}, \quad E = E_A E_B$$

with the rearranged form $E = \frac{x_D}{x_F} \frac{x_F - x_B}{x_D - x_B} \left[1 - \frac{1 - x_D}{1 - x_F} \frac{x_F - x_B}{x_D - x_B} \right]$. The slide’s own worked solution is the graphical one on slide 18: the cumulative-fraction curves for feed, overflow and underflow are plotted together against D_p , and read off at the cut diameter.

Step 1 — mass ratios of overflow and underflow to feed

DERIVED

$$\begin{aligned} \frac{D}{F} &= \frac{x_F - x_B}{x_D - x_B} = \frac{0.47 - 0.195}{0.85 - 0.195} = \frac{0.275}{0.655} = 0.4198 \\ \frac{B}{F} &= \frac{x_F - x_D}{x_B - x_D} = \frac{0.47 - 0.85}{0.195 - 0.85} = \frac{-0.38}{-0.655} = 0.5802 \end{aligned}$$

Mass-balance closure: $D/F + B/F = 0.4198 + 0.5802 = 1.000 \checkmark$

Step 2 — effectiveness of each fraction

DERIVED

$$\begin{aligned} E_A &= \frac{D}{F} \cdot \frac{x_D}{x_F} = 0.4198 \times \frac{0.85}{0.47} = 0.4198 \times 1.8085 = 0.7593 \\ E_B &= \frac{B}{F} \cdot \frac{1 - x_B}{1 - x_F} = 0.5802 \times \frac{0.805}{0.53} = 0.5802 \times 1.5189 = 0.8812 \end{aligned}$$

Step 3 — overall effectiveness

DERIVED

$$E = E_A E_B = 0.7593 \times 0.8812 = 0.6691 \approx 66.9\%$$

The same result from the slide's boxed form:

$$E = \frac{0.85}{0.47} \cdot \frac{0.275}{0.655} \left[1 - \frac{0.15}{0.53} \cdot \frac{0.275}{0.655} \right] = 1.8085 \times 0.4198 \times (1 - 0.1188) = 0.6691$$

VERIFY Both the $E_A E_B$ product and the rearranged boxed form give 0.6691; the component balances also close:

$$\frac{D}{F} x_D + \frac{B}{F} x_B = 0.4198(0.85) + 0.5802(0.195) = 0.470 = x_F \checkmark$$

$$\frac{D}{F}(1 - x_D) + \frac{B}{F}(1 - x_B) = 0.4198(0.15) + 0.5802(0.805) = 0.530 = 1 - x_F \checkmark$$

A 10-mesh screen recovering 76% of the oversize and 88% of the undersize is a healthy result for a commercial screen — consistent with the slide's remark that a commercial screen gives poor separation compared with a testing screen.

FLAG Three notes:

1. The solution assumes the table is to be read exactly *at* the cut size. Strictly, x_F , x_D and x_B should be evaluated on the cumulative curves at $D_p = D_{pc} = 1.651$ mm; the table row at 10 mesh provides those values directly, so no interpolation arises here.
2. The underflow column is blank above 8 mesh (nothing coarse reports to the underflow), which is consistent with $x_B = 0$ at 8 mesh rather than missing data.
3. **The boxed rearrangement on the slide.** Slides 12–15 carry a hand-written derivation in which the constant term of the rearranged expression is scribbled over. This does not affect the answer, because the slides *define* the overall effectiveness as the product of the two direction efficiencies, $E = E_A E_B$; the boxed form evaluates to exactly the same number (0.6691), as verified above, and it also returns the correct limit $E = 1$ for a perfect separation ($x_D = 1$, $x_B = 0$). Both checks are in `sp2-answers-check.py`.

Answers (Q5).

$$\frac{D}{F} = 0.42 \text{ (overflow/feed)}, \quad \frac{B}{F} = 0.58 \text{ (underflow/feed)}.$$

$$E_A = 0.759, \quad E_B = 0.881, \quad \boxed{E = E_A E_B = 0.669 \approx 67\%} \text{ overall effectiveness.}$$

APPENDIX — HOW THESE ANSWERS WERE VERIFIED

Reproducibility

Every **DERIVED** number in this document was produced by `sp2-answers-check.py`, kept beside this file. Running it prints each quantity in the same order as the solutions:

```
python3 sp2-answers-check.py
```

No arithmetic in the solutions was done by hand, so any value can be re-checked by re-running that script or by editing the input data at the top of each block.

What was checked, and against what

Item	Check performed	Result
Q1	$\sum x_i = 1$; the two independent slide forms for $\overline{D}_s$ agree	pass
Q2	every increment diameter equals the mean of its two defining screen openings	pass (13/13, to 0.0005 mm)
Q2	the two routes to N_i/N_T agree; comparison with the published worked solution of this problem	pass (all values)
Q3	slide formula for s read from the slide (image, not OCR); both denominators reported	pass
Q4	ratio route reproduces the direct route; dimensionality of K_R	pass
Q5	$E = E_A E_B$ and the boxed rearranged form agree ($\Delta < 10^{-6}$); both balances close; perfect-separation limit gives $E = 1$	pass

Known limits of this document

- It contains the *five* numericals present in the eight uploaded decks. Slides that merely show a demonstration video, a diagram, or a worked graph were checked and are not treated as questions (see the exclusion list below); no question was invented.
- Where the slides give no method — they never do for the five above — the method was taken from the lecture that set the question, never from outside.
- Answers were re-derived from the data, so if an instructor's answer differs, compare the flagged assumptions first; they are the only places where the sources leave room.

Slides examined and excluded (not questions)

Slide	Why excluded
Mod 1 Lec 5, slides 7–18	hand-written derivations and diagrams, no question posed
Mod 1 Lec 3, slides 3–8	hand-written derivation of average diameters, no question posed
Mod 2 Lec 1, slides 11–15	hand-written derivation of screen effectiveness (source of Q5's method)
Mod 1 Lec 2, slide 13; Lec 1 slide 13; Lec 5 slide 17	data tables/graphs shown as illustration
Mod 2 Lec 2, slides 5–12, 20	diagrams and pressure-gradient profile
Various slides with youtu.be/ youtube.com links	demonstration videos

Provenance summary. Questions, data, formulas and methods: **SOURCE** the uploaded decks. Published-answer confirmation for Q2: **TEXT** the crushed-quartz screen-analysis example of McCabe, Smith & Harriott, *Unit Operations of Chemical Engineering*, Ch. 28 (consulted as a cross-check of the numbers reported above; the method itself is taken from the slides). Everything else — algebra, arithmetic, unit conversions, the checks above: **DERIVED** / **VERIFY** generated. Ambiguities: **FLAG** flagged in place. This file is the editable source; regenerate the PDF after any change.